



CHIPWHISPERER CW308 TARGET

PROACT ASIC Evaluation & Side-Channel Target Board

Hardware reference — chip pinout, signal routing, jumpers, switches, connectors, clock system, Vcore trim, power & current-sense architecture.

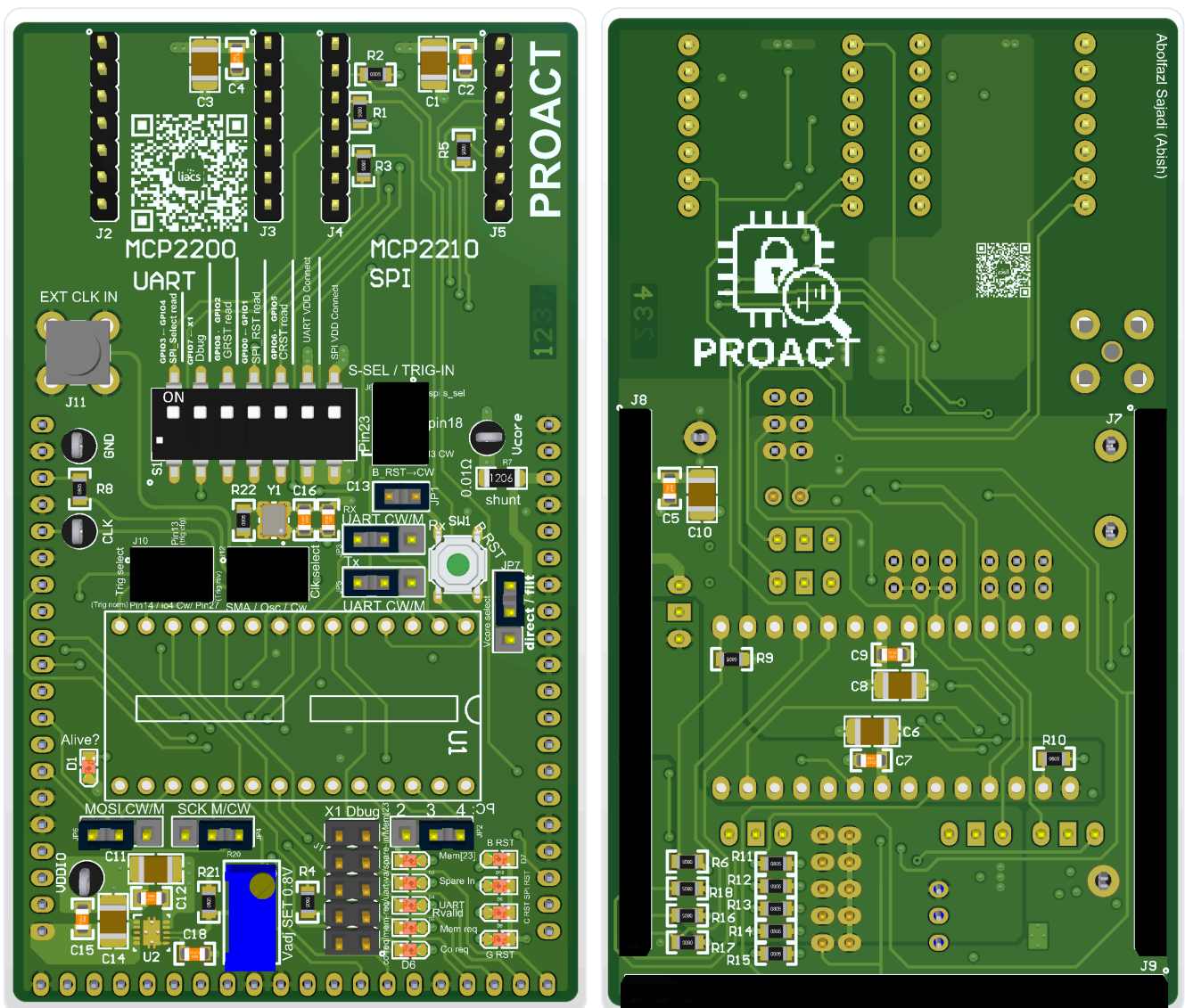
DIP-28 · socket U1

Vcore 0.80–0.90 V trim

3-source clock select

R7 0.01 Ω shunt

CW308 J7 / J8 / J9



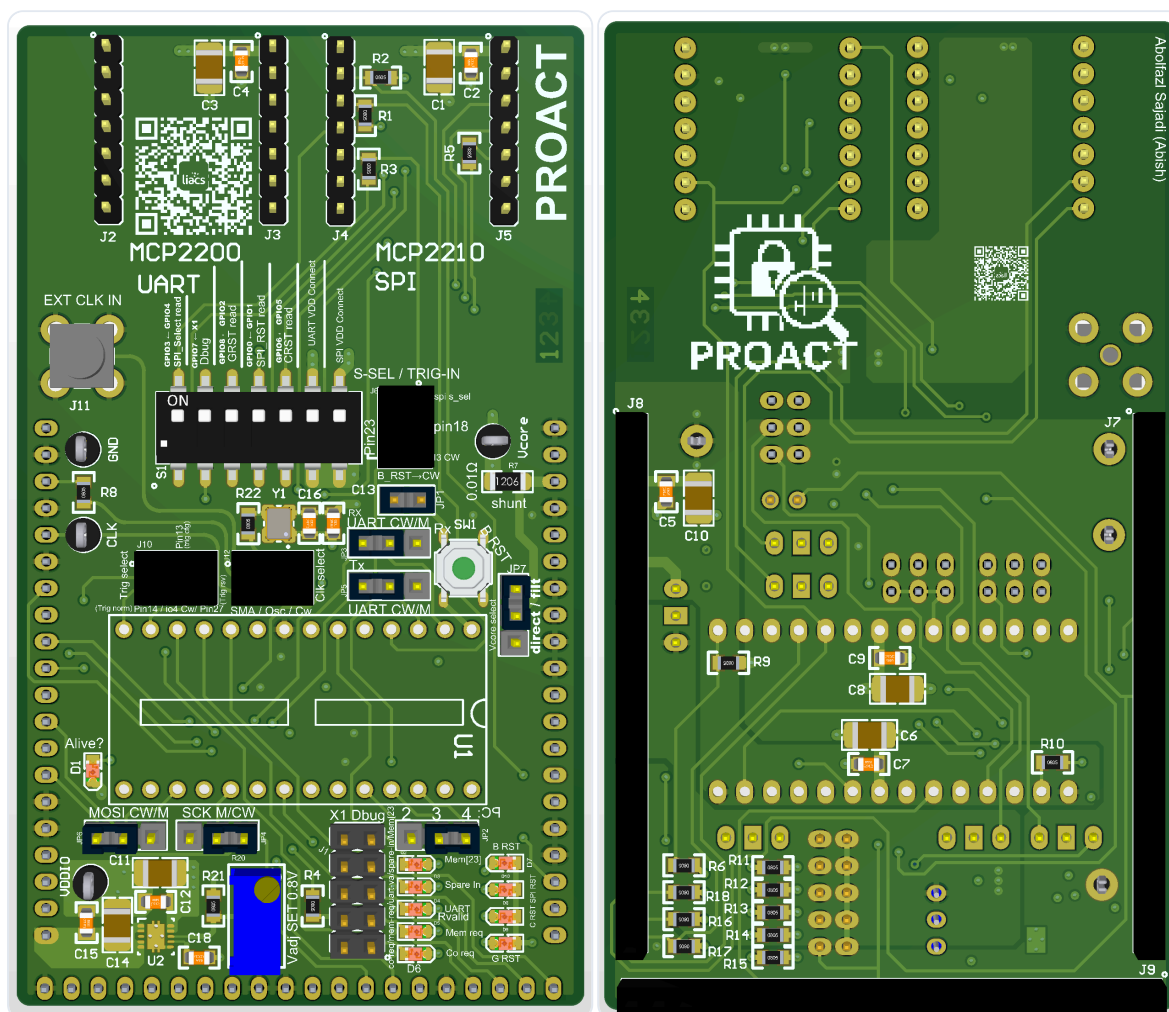
Generated from the board's Altium netlist, placement data and BOM · verified against the design.



PROACT ASIC — Evaluation & Side-Channel Target Board

A ChipWhisperer **CW308 UFO** target board for the **PROACT** secure ASIC. The board hosts the PROACT chip in a DIP-28 socket and lets you drive its UART / SPI / GPIO interfaces either from two on-board **USB bridge modules** (MCP2200 for UART, MCP2210 for SPI + GPIO) or from the ChipWhisperer CW308 platform for power / EM **side-channel analysis** and fault injection — the routing is selected entirely with jumpers. The chip clock can come from an **on-board 50 MHz oscillator**, an **external SMA input**, or the **ChipWhisperer**, and the 0.8 V core rail is **trimmable (0.80 – 0.90 V)** and measured through a 0.01 Ω sense shunt.

[Interactive board wiki](#) — clickable board map with a live jumper simulator · [wiki pages](#) · [PDF reference](#)



At a glance

Target device	PROACT ASIC, 28-pin DIP (socketed at U1)
Host platform	ChipWhisperer CW308 UFO (edge connectors J7 / J8 / J9)
On-board bridges	MCP2200 USB ↔ UART · MCP2210 USB ↔ SPI+GPIO (plug-in modules)
Interfaces	UART, SPI (SIn / sck / SSEL_n), debug outputs, 4 reset lines, 3 trigger outputs + 1 trigger input
Clock	J12 select: on-board Y1 50 MHz (3-4) · external SMA J11 (1-2) · ChipWhisperer (5-6); the chip clock is always echoed to the scope on J7.5 for synchronous sampling

Core supply	TPS74801 LDO, trimmable 0.80 – 0.90 V (R20) — set 0.8 V before inserting the chip; delivered through R7 0.01 Ω sense shunt, optionally via the CW308 L-C filter (JP7)
I/O supply	VDDIO 3.3 V from the CW308 rail
Live monitoring	10 LEDs in 4 color groups (alive · debug · spare-in · resets) + J1 monitor header + CW308 status LEDs
Board size	53.9 mm × 93.3 mm

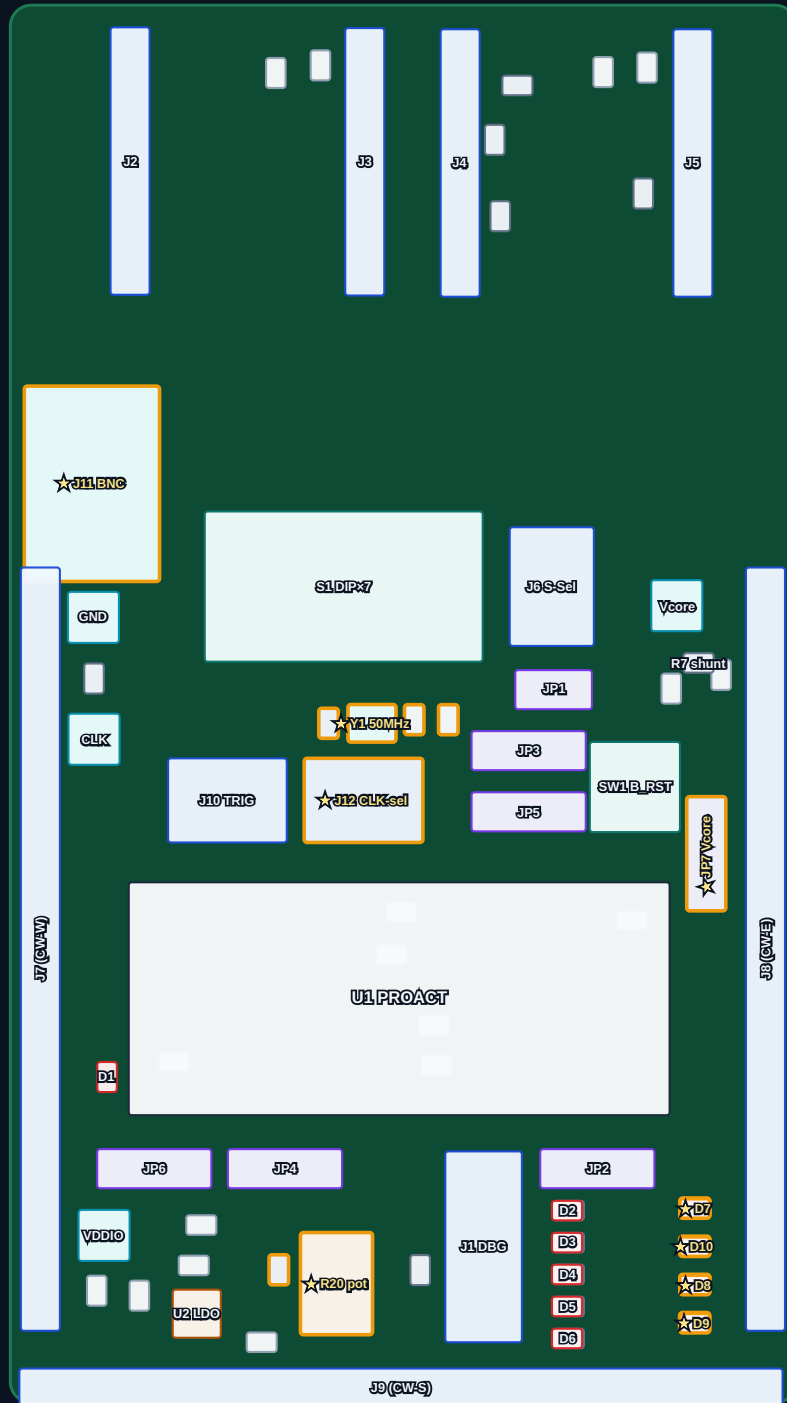
The unifying idea of the board: **every PROACT interface signal can be sourced from the USB bridge module or from the ChipWhisperer**, chosen by a 3-pin jumper whose **centre pin is always the PROACT chip**. Silkscreen tags such as Mosi M/CW and SCK M/CW spell this out — **M** = module, **CW** = ChipWhisperer.

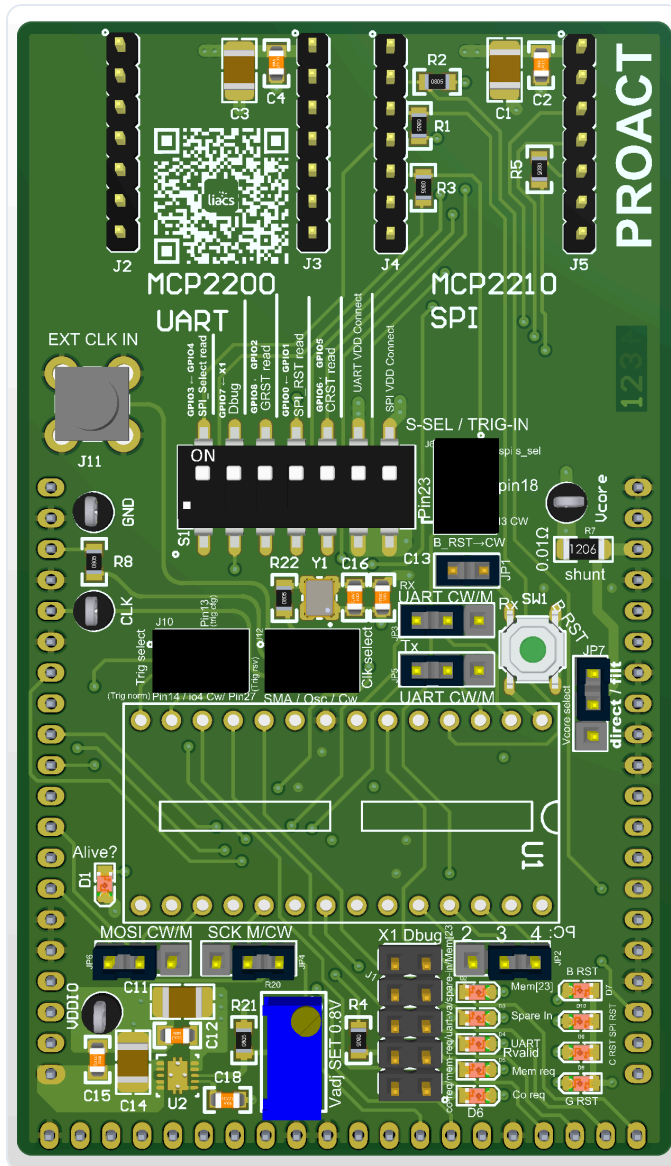
Board tour

Placement view — every component located on the board outline, straight from the final PCB1.PcbDoc placement data.

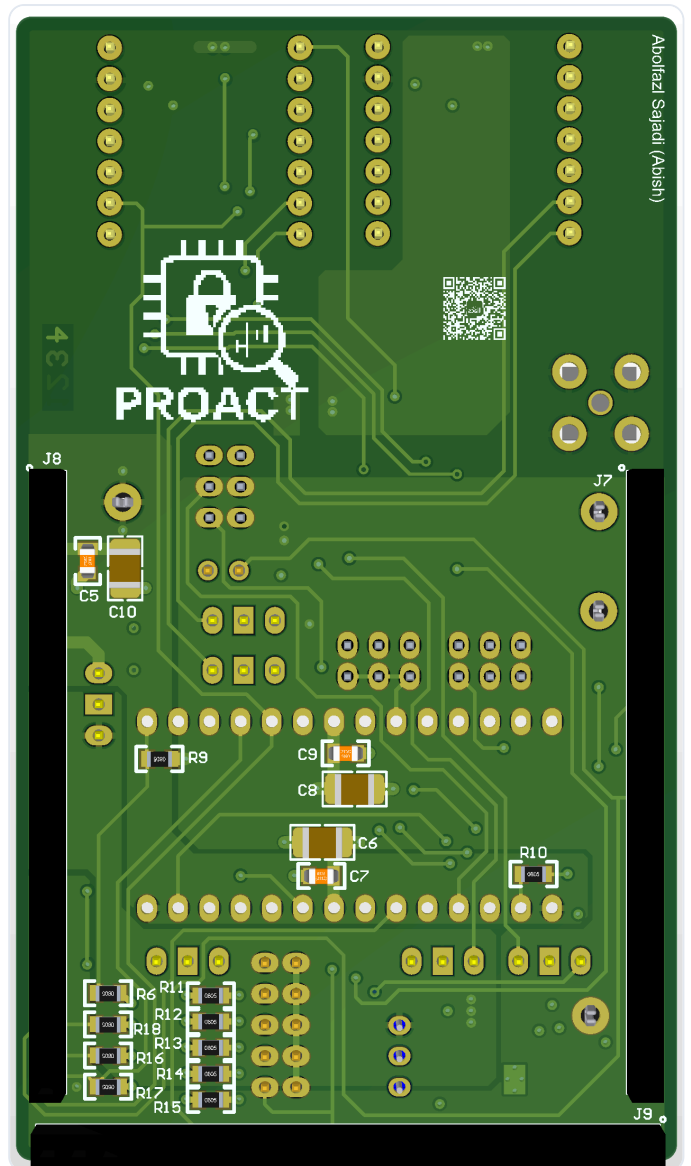
PROACT Board — Placement View

from PCB1.PcbDoc · top view · ★ = clock / Vcore / debug additions



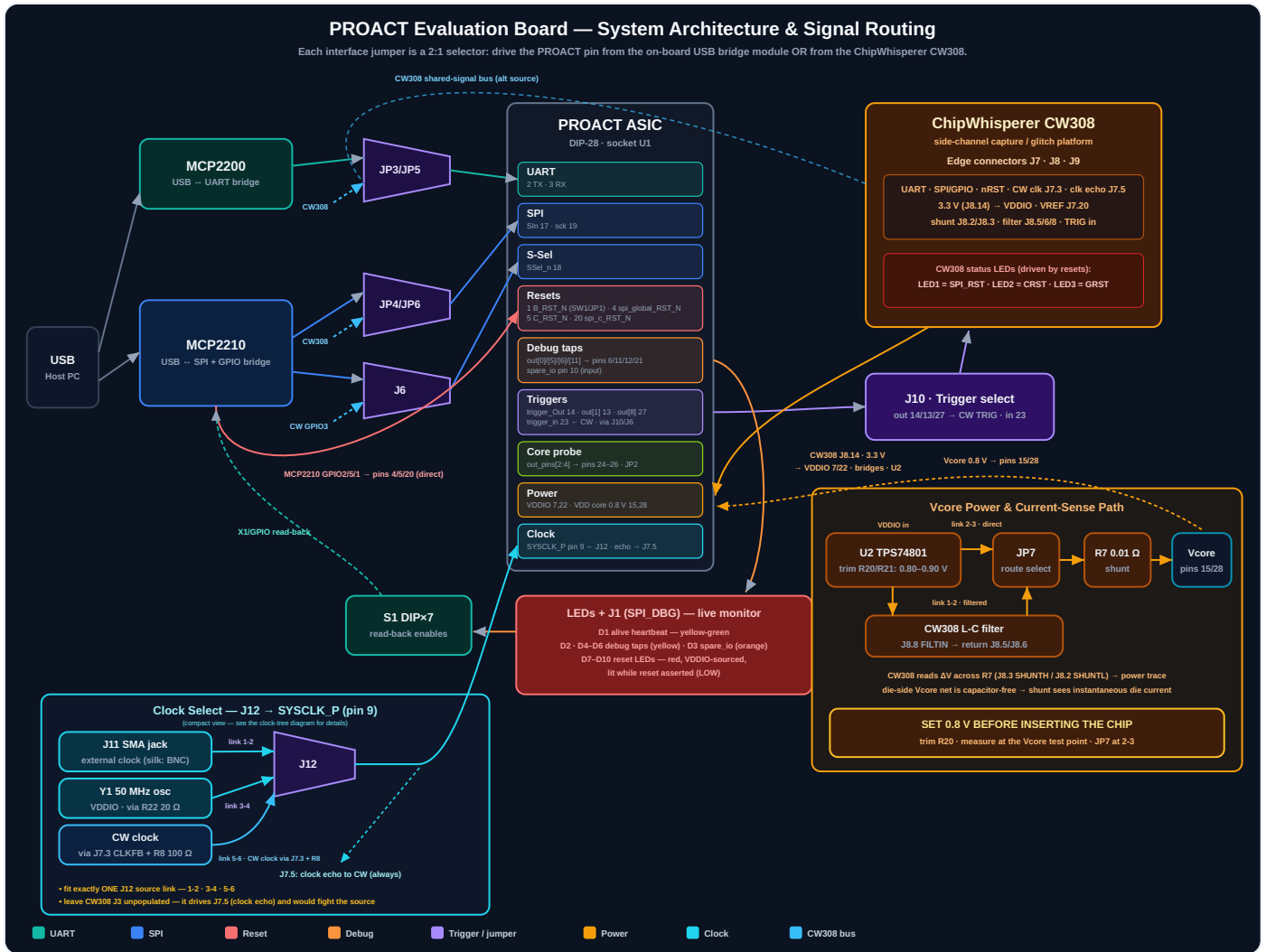


Top side — final board (CAD render, full silkscreen)



Bottom side — PROACT logo & CW308 edge connectors (CAD render)

System architecture & signal routing



Reading the diagram

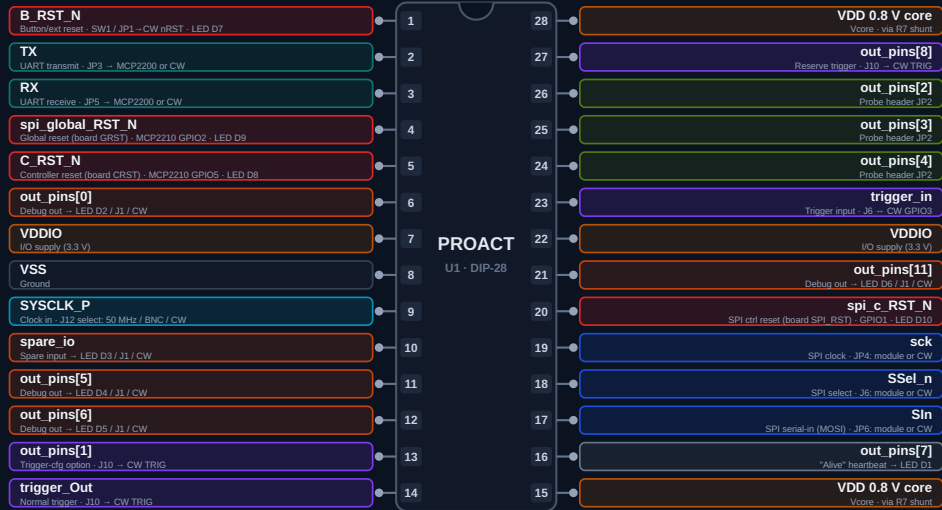
- Two masters, one target.** The PROACT chip in the centre can be talked to by the USB bridge modules (left) or by the ChipWhisperer CW308 (right). The purple jumper "muxes" pick which.
- UART** (JP3 / JP5) routes PROACT TX (pin 2) and RX (pin 3) to the MCP2200 module or the CW308 (J7.8 / J7.7).
- SPI** (JP4 = sck pin 19, JP6 = SIn pin 17) and **S-Sel** (J6 = SSel_n pin 18) route to the MCP2210 module or the CW308.
- Resets** come from MCP2210 GPIO or the button: B_RST_N (pin 1, SW1 / JP1), spi_global_RST_N (pin 4, GPIO2), C_RST_N (pin 5, GPIO5), spi_c_RST_N (pin 20, GPIO1). The three GPIO resets also drive the CW308's status LEDs, and all four light a **red LED (D7–D10) while asserted**.
- Debug outputs** out_pins[0]/[5]/[6]/[11] (pins 6, 11, 12, 21) light the yellow LEDs **D2/D4/D5/D6**, appear on header **J1** and the CW308 south header, and can be read back into the MCP2210 through the **S1** DIP switch. spare_io (pin 10) is a user-drivable input with the orange LED **D3**.
- Triggers** (trigger_out 14, out_pins[1] 13, out_pins[8] 27) are selected on **J10** and fed to the CW308 TRIG input; trigger_in (pin 23) can be driven from CW308 GPIO3 via J6.
- Clock**: SYSCLK_P (pin 9) is fed from the **J12** clock select — on-board 50 MHz Y1, external SMA J11, or the ChipWhisperer (link 5-6) — and is always echoed to the scope on J7.5 for **synchronous sampling**.
- Power / measurement**: CW308 3.3 V → VDDIO; the U2 LDO makes the trimmable 0.8 V core rail, which reaches the chip through **JP7** (direct or via the CW308 filter) and the **R7** shunt; the CW308 measures the drop across R7 to capture the power trace.

PROACT chip pinout (DIP-28)

Socket **U1**. Pin 1 is top-left with the package notch up. Colours group pins by function. IN · PU = input with internal pull-up.

PROACT 22FDX — DIP-28 Pinout

socket U1 · pin 1 top-left, notch up · RTL names from pins.png · board nets verified vs netlist



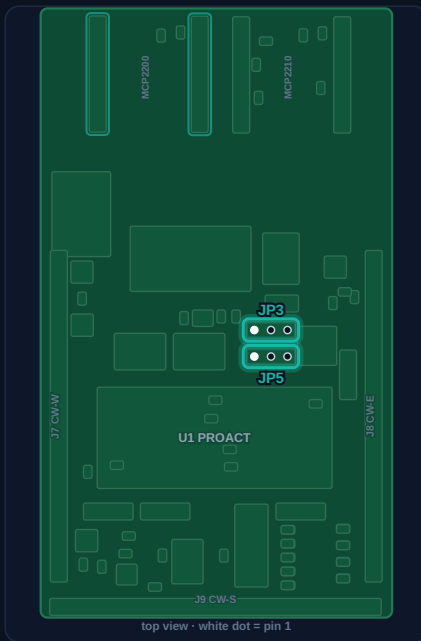
■ Power
 ■ Ground
 ■ Clock
 ■ UART
 ■ SPI
 ■ Reset
 ■ Trigger
 ■ Debug monitor
 ■ Core control
 ■ Status

Pin	Signal	Dir	Function	Routing / destination
1	B_RST_N	IN-PU	Button / external reset (active low)	SW1 ; R9 10 k pull-up; JP1 → CW308 nRST ; red LED D7
2	TX	OUT	UART transmit	JP3 → MCP2200 RX or CW308 (J7.8)
3	RX	IN	UART receive	JP5 → MCP2200 TX or CW308 (J7.7)
4	spi_global_RST_N	IN-PU	Global reset (board net GRST)	MCP2210 GPIO2; CW308 LED3; red LED D9
5	C_RST_N	IN-PU	Controller reset (board net CRST)	MCP2210 GPIO5; CW308 LED2; red LED D8
6	out_pins[0]	OUT	Debug output	yellow LED D2 · J1.9 · J9.20
7	VDDIO	PWR	I/O supply	3.3 V
8	VSS	GND	Ground	—
9	SYSCLK_P	IN	Chip clock	from J12 clock select · CLK test point · J7.5
10	spare_io	IN	Spare input (user-driven)	orange LED D3 · J1.7 · J9.19
11	out_pins[5]	OUT	Debug output	yellow LED D4 · J1.5 · J9.18
12	out_pins[6]	OUT	Debug output	yellow LED D5 · J1.3 · J9.17
13	out_pins[1]	OUT	Trigger (config)	J10 (3-4) → CW308 TRIG
14	trigger_Out	OUT	Trigger (normal)	J10 (2-4) → CW308 TRIG
15	VDD 0.8 V core	PWR	Core supply	via R7 shunt (with pin 28)
16	out_pins[7]	OUT	Alive heartbeat	yellow-green LED D1 (blinks = alive)
17	SIn	IN	SPI data in (MOSI)	JP6 → MCP2210 or CW308
18	SSel_n	IN	SPI select (active low)	J6 → MCP2210 GPIO4 or CW308 GPIO3
19	sck	IN	SPI clock	JP4 → MCP2210 or CW308
20	spi_c_RST_N	IN-PU	SPI-controller reset (board net SPI_RST)	MCP2210 GPIO1; CW308 LED1; red LED D10
21	out_pins[11]	OUT	Debug output	yellow LED D6 · J1.1 · J9.11
22	VDDIO	PWR	I/O supply	3.3 V
23	trigger_in	IN	Trigger input	J6.6 ← CW308 GPIO3 (link J6 5-6)
24	out_pins[4]	OUT	IBEX PC probe (bit 4)	JP2.3
25	out_pins[3]	OUT	IBEX PC probe (bit 3)	JP2.2
26	out_pins[2]	OUT	IBEX PC probe (bit 2)	JP2.1

JP3 + JP5 — UART routing

PROACT eval board · jumper card

two 1x3 headers · center pin = chip UART · 1-2 = ChipWhisperer · 2-3 = MCP2200 USB module



1-2 · CW

JP3 · UART-TX



chip TX → CW308 TIO2

JP5 · UART-RX



CW308 TIO1 → chip RX

Chip UART talks to the ChipWhisperer capture board (SimpleSerial).

on CW308

2-3 · MODULE

JP3 · UART-TX



chip TX → module RX

JP5 · UART-RX



module TX → chip RX

Chip UART talks to the MCP2200 USB-UART module (J2/J3 socket).

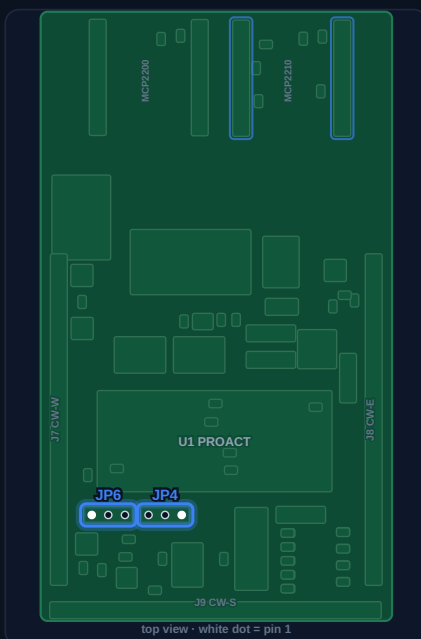
standalone · USB serial

Move JP3 and JP5 TOGETHER — always the same side.

JP4 + JP6 — SPI SCK / MOSI routing

PROACT eval board · jumper card

two 1x3 headers · center pin = chip SPI input · 1-2 = ChipWhisperer · 2-3 = MCP2210 USB module



1-2 · CW

JP4 · SPI clock



CW308 → chip SCK

JP6 · SPI data in



CW308 → chip MOSI

SPI master = ChipWhisperer (loads keys / configuration).

on CW308

2-3 · MODULE

JP4 · SPI clock



MCP2210 → chip SCK

JP6 · SPI data in



MCP2210 → chip MOSI

SPI master = MCP2210 USB-SPI module (J4/J5 socket).

standalone · USB SPI

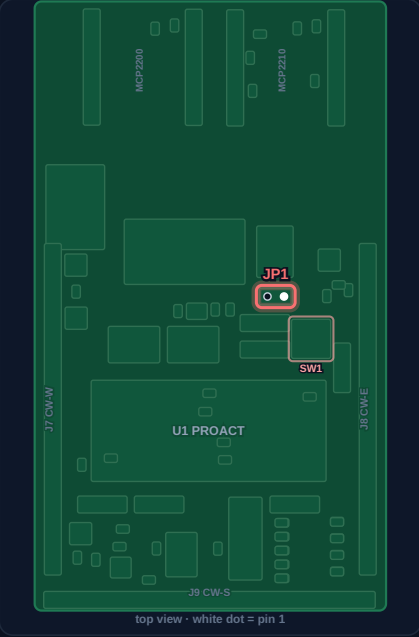
Move JP4 and JP6 TOGETHER · the S-Sel driver is chosen on J6.

Two-pin, probe and special jumpers

Jumper	Type	Function
JP1	1x2	Close to tie PROACT <code>B_RST_N</code> (pin 1) to the CW308 <code>nRST</code> line. Leave open to reset only from <code>Sw1</code> .
JP2	1x3	Probe header for IBEX PC bits — <code>out_pins[2]/[3]/[4]</code> (chip pins 26/25/24). No jumper fitted.
JP7	1x3	Vcore route: 1-2 = core rail through the CW308 L-C filter · 2-3 = direct from the LDO. See Core voltage .
J12	2x3	Clock source select (1-2 SMA · 3-4 osc · 5-6 ChipWhisperer). See Clock system .

JP1 — Reset from ChipWhisperer

1x2 header · closed = CW308 nRST line reaches the chip reset



CLOSED

1

2

B_RST_N

CW nRST

chip 1 · SW1 · R9

J7.11 / J9.15

CW308 nRST → chip B_RST_N

ChipWhisperer (J7.11 / J9.15) can reset the chip. SW1 push-button still works.

✓ board mounted on CW308

OPEN

1

2

B_RST_N

CW nRST

shunt parked

chip 1 · SW1 · R9

J7.11 / J9.15

SW1 only → chip B_RST_N

Reset only from push-button SW1. R9 10 k holds B_RST_N high to VDDIO.

standalone bench use

JP2 — Probe header (no jumper)

1x3 header · core debug bits out_pins[2..4] · probe only — there is no jumper logic here



PROBE HEADER — DO NOT FIT A JUMPER

1

2

3

out[2]

out[3]

out[4]

U1.26

U1.25

U1.24

chip out_pins[2..4] → scope / logic analyzer

Clip an oscilloscope or logic-analyzer probe here to watch the core debug/PC bits during operation.

Fitting a shunt would short two chip outputs together.

J6 — S-Sel & trigger-input block (2x3): selects the source of PROACT `Ssel_n` (pin 18) and can route the PROACT trigger input (pin 23) to the CW308.

Link on J6	Effect
1-3	MCP2210 GPIO4 drives <code>Ssel_n</code> (pin 18) — SPI select from the bridge module
3-5	CW308 GPIO3 (J7.9) drives <code>Ssel_n</code> (pin 18) — SPI select from the ChipWhisperer
5-6	CW308 GPIO3 drives <code>trigger_in</code> (pin 23)

J10 — trigger select (2x3): pin 4 is the CW308 `GPIO4/TRIG` line (J7.10); jumper it to one PROACT trigger source.

PROACT Evaluation Board — Reference

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Link on J10	Trigger source → CW308 TRIG
2-4	trigger_Out — normal trigger (pin 14)
3-4	out_pins[1] — config trigger (pin 13)
4-6	out_pins[8] — reserve / software trigger (pin 27)

J6 — S-Sel source & trigger-in

2x3 header · vertical link on pins 1-3-5 picks the S-Sel driver · horizontal link 5-6 routes the CW trigger-in

PROACT eval board · jumper card

1-3 · MODULE drives S-Sel

MCP2210 GPIO4 · J4.5 → 1
chip S-Sel_n · U1.18 → 3
CW GPIO3 · J7.9 → 5

2 NC
4 NC
6 chip trig-in · U1.23

MCP2210 GPIO4 → chip S-Sel_n
USB-SPI module selects the chip.
✓ with USB module

3-5 · CW drives S-Sel

MCP2210 GPIO4 · J4.5 → 1
chip S-Sel_n · U1.18 → 3
CW GPIO3 · J7.9 → 5

2 NC
4 NC
6 chip trig-in · U1.23

CW308 GPIO3 → chip S-Sel_n
ChipWhisperer selects the chip.

5-6 · CW trigger-in

MCP2210 GPIO4 · J4.5 → 1
chip S-Sel_n · U1.18 → 3
CW GPIO3 · J7.9 → 5

2 NC
4 NC
6 chip trig-in · U1.23

CW308 GPIO3 → chip trig-in
CW pulses the chip trigger input.
Combinable with 1-3 (not 3-5).

Pins 2 and 4 are not connected.

J10 — Trigger-out select

2x3 header · choose which chip trigger reaches the CW308 TRIG / GPIO4 line (J7.10)

PROACT eval board · jumper card

2-4 · NORMAL

NC → 1
out[1] config · U1.13 → 3
NC → 5

2 trigger_Out · U1.14
4 CW TRIG · J7.10
6 out[8] reserve · U1.27

trigger_Out → CW TRIG
Main encryption trigger (U1.14).
✓ default

3-4 · CONFIG

NC → 1
out[1] config · U1.13 → 3
NC → 5

2 trigger_Out · U1.14
4 CW TRIG · J7.10
6 out[8] reserve · U1.27

out_pins[1] → CW TRIG
Configuration-phase trigger (U1.13).

4-6 · RESERVE

NC → 1
out[1] config · U1.13 → 3
NC → 5

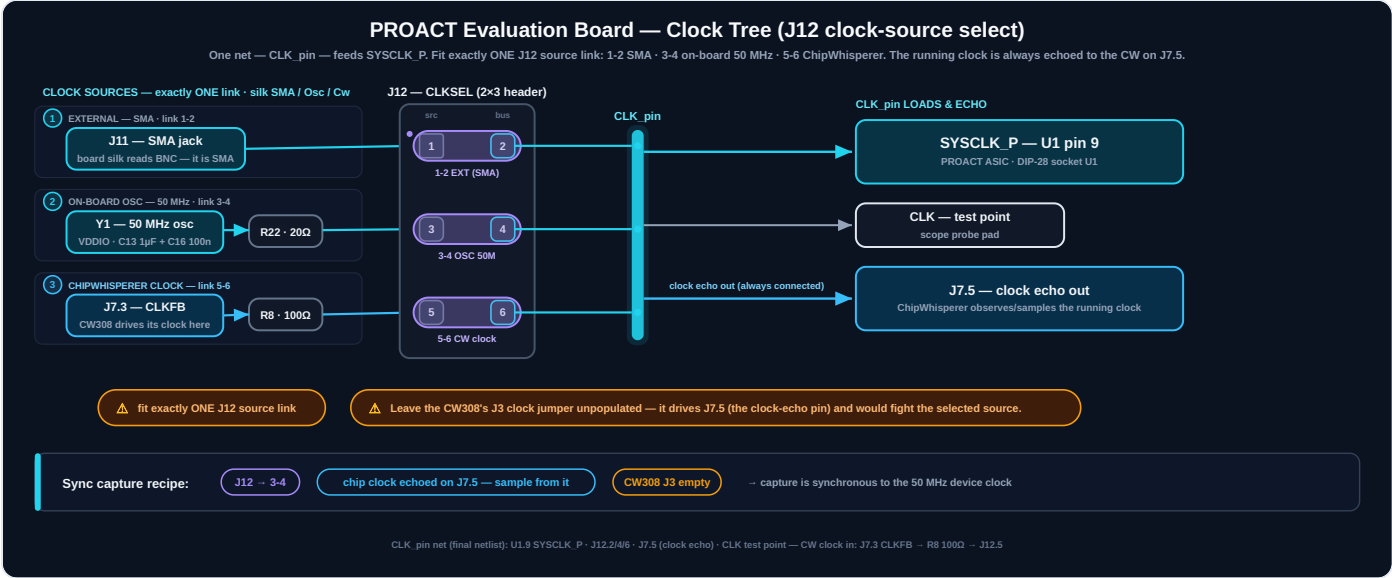
2 trigger_Out · U1.14
4 CW TRIG · J7.10
6 out[8] reserve · U1.27

out_pins[8] → CW TRIG
Reserve trigger output (U1.27).

Pins 1 and 5 are not connected · fit exactly one link.

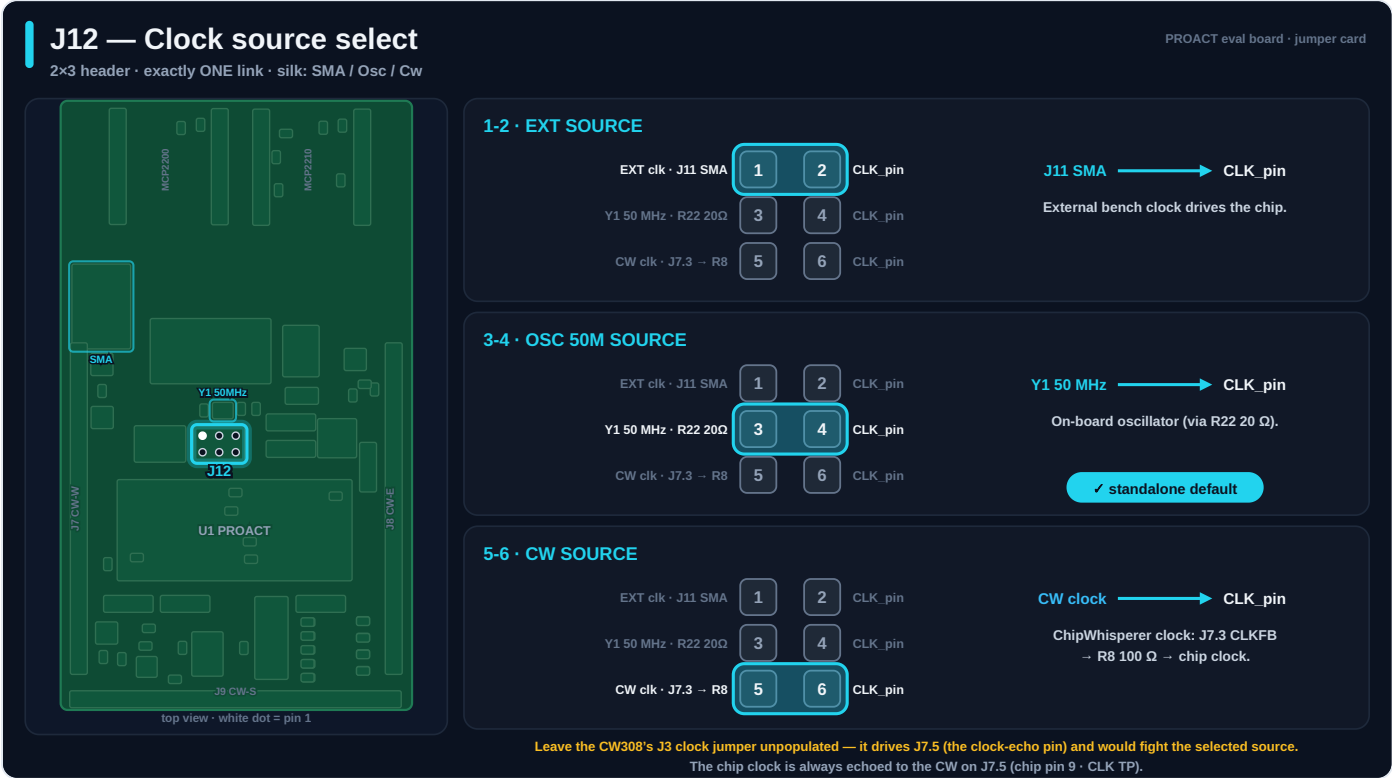
Clock system

The chip clock `SYSCLK_P` (pin 9) can come from **three sources, selected on J12** (board silk: `Clk select — SMA / Osc / Cw`), and is **always echoed to the ChipWhisperer on J7.5** so the scope can sample synchronously with the target clock.



Source	J12 link	Path
On-board 50 MHz	3-4	Y1 oscillator → R22 20 Ω → chip clock
External input	1-2	J11 coaxial jack → chip clock. <i>The fitted connector is SMA (the board silk reads "BNC").</i>
ChipWhisperer	5-6	CW clock arriving on J7.3 (CW308 CLKFB line) → R8 100 Ω → chip clock

Clock echo — synchronous sampling. The chip clock is **permanently wired to J7.5**, so the ChipWhisperer side can always observe the running clock and lock its sampling to it — whatever the selected source, with no extra jumper.

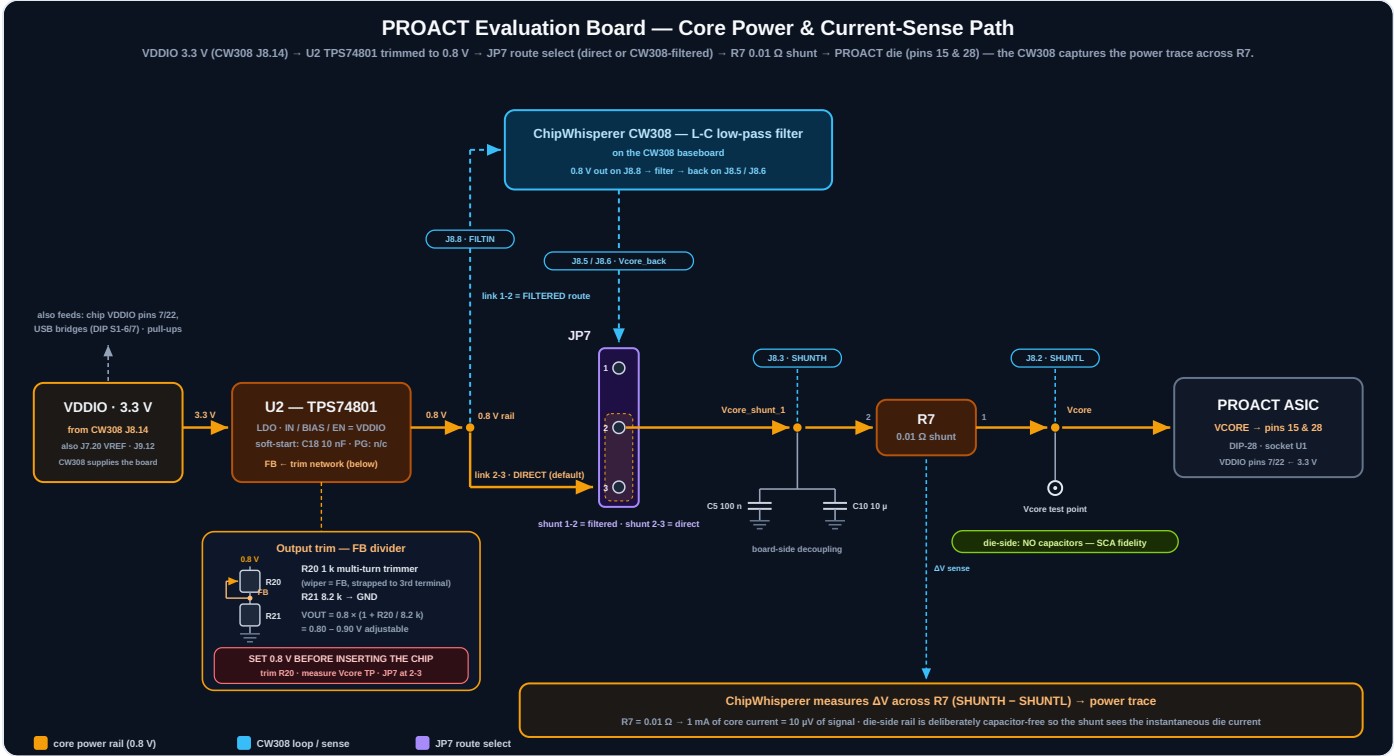


Rules - Fit exactly one source link on J12 (1-2 , 3-4 or 5-6). - Leave the CW308's J3 clock jumper unpopulated — it drives J7.5 (the clock-echo pin) and would fight the selected source. - The CLK test point sits directly on the chip clock net for probing.

Core voltage (Vcore): set 0.8 V before inserting the chip

The 0.8 V core rail is generated on-board by the U2 TPS74801 LDO EXT and trimmed with the multi-turn potentiometer R20:

$V_{core} = 0.8\text{ V} \times (1 + R20 / 8.2\text{ k}\Omega) \rightarrow \text{adjustable } 0.80 - 0.90\text{ V}$



⚠ Do this before the chip ever meets the board: 1. Leave the PROACT chip out of socket U1 . 2. Set JP7 to **2-3** (direct). 3. Apply 3.3 V VDDIO (mount on the CW308, or feed the VDDIO test point on the bench). 4. Meter between the **Vcore test point** and GND (no load → no shunt drop). 5. Turn R20 (multi-turn) until the meter reads **0.800 V**. 6. Power down, insert the chip (notch up), power up and re-check under load.

JP7 — core-rail route

Link	Route	When to use
1-2	0.8 V → J8.8 (FILTIN) → CW308 L-C low-pass filter → back on J8.5/6 → shunt	Side-channel capture on the CW308 — the filter cleans the rail so the shunt sees the die, not supply noise
2-3	0.8 V → shunt, direct	Bench use off the CW308, or when you want the shortest supply path

JP7 — Vcore routing

1×3 header · 1-2 = core rail through the CW308 L-C filter · 2-3 = direct from on-board LDO

top view · white dot = pin 1

1-2 · FILTERED

1

2

3

CW filter
J8.5 / J8.6

R7 shunt
→ core 15/28

U2 LDO
0.8 V

CW308 filter → R7 → core

0.8 V leaves the board on J8.8 (FILTIN), passes the CW308 L-C low-pass filter, returns on J8.5/6 and feeds the core through the 0.01 Ω measurement shunt R7.

✓ SCA capture — board on CW308

2-3 · DIRECT

1

2

3

CW filter
J8.5 / J8.6

R7 shunt
→ core 15/28

U2 LDO
0.8 V

U2 LDO 0.8 V → R7 → core

On-board LDO U2 feeds the core directly through shunt R7 — CW308 filter bypassed. Set 0.8 V on R20 before inserting the chip.

bench / standalone use

The die side of the shunt (net `Vcore`, chip pins `15` and `28`) is **deliberately capacitor-free**: decoupling sits *before* the shunt (`C5` / `C10`), so the instantaneous die current flows through `R7` and appears in the power trace.

S1 DIP switch (read-back & power enables)

`S1` is a 7-position DIP switch. Switches `1–5` enable GPIO **read-back** loops (so the USB host can read the state of a signal it — or the chip — is driving); switches `6–7` power the two USB bridge modules.

Switch	Enables	Purpose
1	MCP2210 GPIO3 reads GPIO4	Read back the <code>Ssel_n</code> state
2	MCP2210 GPIO7 reads <code>x1</code>	Read back the debug signal selected on <code>J1</code>
3	MCP2210 GPIO8 reads GPIO2	Read back the global reset (<code>spi_global_RST_N</code>)
4	MCP2210 GPIO0 reads GPIO1	Read back the SPI reset (<code>spi_c_RST_N</code>)
5	MCP2210 GPIO6 reads GPIO5	Read back the controller reset (<code>C_RST_N</code>)
6	MCP2200 <code>VDD</code> → 3.3 V	Power the UART bridge module
7	MCP2210 <code>VDD</code> → 3.3 V	Power the SPI bridge module

Connectors & headers

Ref	Type	Role
U1	DIP-28 socket	PROACT ASIC
J1	2×5 male (<code>SPI_DBG</code>)	Debug-signal monitor (see below)
J2 / J3	1×7 male ×2	MCP2200 USB-UART module socket
J4 / J5	1×7 male ×2	MCP2210 USB-SPI module socket
J6	2×3 female	S-Sel / trigger-input routing
J10	2×3 female	Trigger select
J11	coaxial jack	External clock input — SMA fitted (silk reads "BNC")
J12	2×3 female	Clock source select (SMA / osc / CW)

PROACT Evaluation Board — Reference

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Ref	Type	Role
J7	1×20 female	CW308 West edge connector (clock · UART · SPI · GPIO · nRST)
J8	1×20 female	CW308 East edge connector (power rails · shunt sense · filter loop · status LEDs)
J9	1×20 female	CW308 South edge connector (debug taps · SPI · nRST)

J1 — SPI_DBG signal monitor (2×5)

Each **odd** pin is a live PROACT debug signal; the adjacent **even** pin is the common X1 line. Fit a jumper across a row to route that signal onto X1 , which the MCP2210 reads via **GPIO7** (enable S1-2). You can equally probe the odd pins directly.

J1 pin	Signal	PROACT pin	LED
9	out_pins[0]	6	D2
7	spare_io	10	D3
5	out_pins[5]	11	D4
3	out_pins[6]	12	D5
1	out_pins[11]	21	D6
2,4,6,8,10	X1 common (→ MCP2210 GPIO7)	—	—

J7 — CW308 West connector (clock & control)

J7 pin	Net	Meaning
3	CLKFB	ChipWhisperer clock in — selected by J12 5-6, through R8 100 Ω
5	CLKIN	Chip clock echo out to the ChipWhisperer (always connected — keep CW308 J3 unpopulated)
7	TI01	UART toward chip RX (via JP5 1-2)
8	TI02	Chip TX toward the scope (via JP3 1-2)
9	GPIO3	S-Sel / trigger-in source (via J6)
10	GPIO4 / TRIG	Trigger line (via J10)
11	nRST	Reset line (via JP1)
12 / 14	SCK / MOSI	SPI from the ChipWhisperer (via JP4 / JP6 1-2)
20	VREF	Level reference = board VDDIO

J8 — CW308 East connector (power & measurement)

J8 pin	Net	Meaning
2	Vcore (SHUNT-L)	Die side of the R7 sense shunt
3	Vcore_shunt_1 (SHUNT-H)	Supply side of the shunt
5, 6	Vcore_back	Filtered 0.8 V returning from the CW308 L-C filter (used when JP7 = 1-2)
8	0.8 V (FILTIN)	Core rail from the U2 LDO into the CW308 filter (used when JP7 = 1-2)
11	1.2 V	CW308 rail (unused)
12	1.8 V	CW308 rail (unused)
13	2.5 V	CW308 rail (unused)
14	VDDIO (3.3 V)	I/O supply into the board
15	5 V	CW308 rail (unused)
18	SPI_RST → LED1	CW308 status LED
19	CRST → LED2	CW308 status LED
20	GRST → LED3	CW308 status LED

J9 — CW308 South connector (debug taps)

J9 pin	Net
11	out_pins[11] (pin 21)
12	VDDIO

J9 pin	Net
13	sck — CW side of JP4
14	SIn — CW side of JP6
15	nRST (with JP1)
17	out_pins[6] (pin 12)
18	out_pins[5] (pin 11)
19	spare_io (pin 10)
20	out_pins[0] (pin 6)

LEDs

Ten on-board LEDs in **four color groups**:

LED(s)	Color	Signal (chip pin)	Lights when
D1	 yellow-green	out_pins[7] — alive (16)	the heartbeat output is high — blinking = chip alive
D2 / D4 / D5 / D6	 yellow	out_pins[0]/[5]/[6]/[11] (6 / 11 / 12 / 21)	the debug output is high
D3	 orange	spare_io (10)	you drive the spare input high (it is an input — the chip never lights it)
D7 / D8 / D9 / D10	 red	B_RST_N (1) / C_RST_N (5) / spi_global_RST_N (4) / spi_c_RST_N (20)	the reset is asserted (line low)

 **All four red LEDs on at once is normal during reset — it is not a fault.** They are wired from VDDIO through the LED into the reset line, so a low (active) reset lights them.

 **Matching the board print:** the silkscreen labels the LEDs as D1 "Alive?", D2 "Mem[23]", D3 "Spare In", D4 "UART Rvalid", D5 "Mem req", D6 "Co req"; the reset LED block reads B_RST · SPI_RST · C_RST · G_RST .

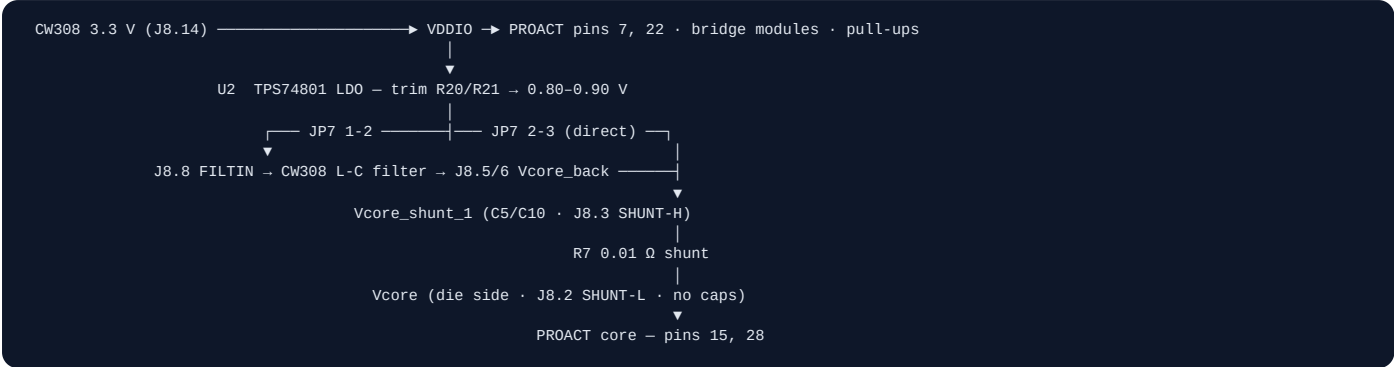
CW308 motherboard status LEDs (driven by the reset lines through J8.18/19/20):

CW308 LED	Signal
LED1	spi_c_RST_N — SPI reset
LED2	C_RST_N — controller reset
LED3	spi_global_RST_N — global reset

Test points

Test point	Net	Use
Vcore	Core rail, die side	Trim target for the 0.8 V procedure; probe high-impedance only — this node is deliberately capacitor-free
VDDIO	3.3 V I/O rail	Probe / feed the I/O supply
CLK	Chip clock (pin 9)	Observe the selected clock
GND	Ground	Scope / meter reference
R7	Shunt	0.01 Ω current-sense resistor with two probe holes for a differential probe

Power & current-sense architecture



- **VDDIO (3.3 V)** comes from the CW308 rail (J8.14) and supplies the PROACT I/O ring, the USB bridge modules (via S1-6/7), the pull-ups and the U2 LDO. For USB-only bench use off the ChipWhisperer, feed 3.3 V into the VDDIO test point instead.
- **Vcore** is trimmed with R20 ([procedure above](#)) and routed by JP7 — through the CW308 filter for capture, or direct for the bench.
- **Measurement.** R7 is 0.01 Ω, so 1 mA of core current is 10 μV across the shunt — use the ChipWhisperer low-noise amplifier on the MEAS output, or a differential probe directly in the two R7 probe holes.
- Decoupling: 10 μF + 100 nF pairs on VDDIO and on the supply side of the shunt; **none** on the die side (by design, for side-channel fidelity).

ChipWhisperer CW308 setup

CW308 motherboard settings for this board

CW308 control	Set to	Why
3.3 V rail	On	Supplies VDDIO via J8.14 .
Filter input	Victim-supplied (FILTIN)	With JP7 = 1-2 the 0.8 V rail from U2 goes through the CW308 L-C filter and back. Do not drive it from VADJ . With JP7 = 2-3 the filter is out of the loop.
J3 clock jumper	Unpopulated	J3 drives J7.5 , which carries the board's clock echo — fitting it would fight the selected clock source. The CW-as-source option is J12 5-6, not J3 .
VREF	From victim	Uses this board's VDDIO (3.3 V, J7.20) as the level reference.
MEAS SMA → Capture	Connect	Feeds the shunt voltage into the ChipWhisperer ADC (or a scope).

ChipWhisperer software notes

- **UART direction is mirrored vs. the CW default:** the chip's TX arrives on TIO2 and the chip's RX is driven from TIO1 — configure tio1 = serial_tx , tio2 = serial_rx .
- **Synchronous sampling:** the chip clock is always available to the ChipWhisperer side on J7.5 — sample from it for phase-locked traces.

Board-side settings for capture

- JP3 JP5 JP4 JP6 → CW (1-2) so UART/SPI come from the ChipWhisperer.
- J6 → 3-5 (CW308 GPIO3 drives SSEL_n); add 5-6 to drive trigger_in instead/as needed.
- J10 → pick the trigger fed to TRIG (2-4 normal · 3-4 config · 4-6 reserve).
- J12 → pick the clock source (see [Clock system](#)).
- JP7 → 1-2 (filtered) for capture.
- JP1 → closed if you want the CW308 nRST to reset the chip.
- S1-6 / S1-7 → off (the USB bridges stay idle and unpowered during capture).
- **Precondition:** Vcore already trimmed to 0.800 V.

USB bridge modules

Two Microchip USB-bridge break-out modules plug into the 1×7 header pairs.

MCP2210 — USB ↔ SPI + GPIO (sockets J4 / J5)

Module pin	Signal	Connected to
1	GPIO0	reads GPIO1 (spi_c_RST_N) when S1-4 on
2	GPIO1	SPI_RST → PROACT pin 20 (spi_c_RST_N)
3	GPIO2	GRST → PROACT pin 4 (spi_global_RST_N)
4	GPIO3	reads GPIO4 (SSeL_n) when S1-1 on
5	GPIO4	SSeL_n → J6 (1-3) → PROACT pin 18
6	MOSI	JP6 (2-3) → PROACT pin 17 (SIn)
7	SCK	JP4 (2-3) → PROACT pin 19 (sck)
8	MISO	not connected
9	GPIO5	CRST → PROACT pin 5 (c_RST_N)
10	GPIO6	reads GPIO5 (c_RST_N) when S1-5 on
11	GPIO7	reads X1 debug bus (J1) when S1-2 on
12	GPIO8	reads GPIO2 (spi_global_RST_N) when S1-3 on
13	GND	ground
14	VDD	3.3 V when S1-7 on

MCP2200 — USB ↔ UART (sockets J2 / J3)

Module pin	Signal	Connected to
6	TX	JP5 (2-3) → PROACT RX (pin 3)
7	RX	JP3 (2-3) → PROACT TX (pin 2)
14	VDD	3.3 V when S1-6 on

Typical configurations

A. USB bench bring-up (talk to PROACT from a PC, no ChipWhisperer)

Setting	Value
Precondition	Vcore trimmed to 0.800 V (procedure)
S1-6 , S1-7	on (power both bridge modules)
S1-1...5	on as needed for GPIO read-back
JP3 , JP5 , JP4 , JP6	M (2-3)
J6	1-3 (MCP2210 GPIO4 → SSeL_n)
J12	3-4 (on-board 50 MHz)
JP7	2-3 (direct)
JP1	open (reset via SW1)
VDDIO	feed 3.3 V into the VDDIO test point

B. ChipWhisperer capture, CW-clocked

Setting	Value
Mount board on the CW308	—
JP3 , JP5 , JP4 , JP6	CW (1-2)
J6	3-5 (CW GPIO3 → SSeL_n)
J10	choose trigger: 2-4 normal · 3-4 config · 4-6 reserve
J12	5-6 (clock from the ChipWhisperer, via J7.3 / R8)
CW308 J3	unpopulated
JP7	1-2 (through the CW308 filter)

Setting	Value
JP1	closed (reset from CW308 nRST) if desired
S1-6 , S1-7	off

C. Synchronous capture from the on-board oscillator (flagship SCA setup)

Setting	Value
As configuration B, except:	
J12	3-4 (the on-board Y1 clocks the chip)
Scope clock	sample from the clock echoed on J7.5 for phase-locked traces

Bill of materials

27 line items (final BOM, JLCPCB part numbers).

Ref(s)	Value	Description	Footprint	JLCPCB	Note
C1, C3, C6, C8, C10, C11, C14	10 μF	Capacitor	1210	C77100	
C2, C4, C5, C7, C9, C12, C15, C16	100 nF	Capacitor	0805	C1711	
C13	1 μF	Capacitor (Y1 decoupling)	0805	C1848	
C18	10 nF	Capacitor (LDO soft-start)	0805	C1710	
D1	yellow-green	LED — alive heartbeat	0805	C84257	
D2, D4, D5, D6	yellow	LED — debug outputs	0805	C84261	
D3	orange	LED — spare_io	0805	C84262	
D7–D10	red	LED — resets	0805	C84256	
J1	—	Header, male 2×5	2.54 mm	C492422	
J2, J3, J4, J5	—	Header, male 1×7	2.54 mm	C124418	
J6, J10, J12	—	Header, female 2×3	2.54 mm	C65114	
J7, J8, J9	—	Header, female 1×20	2.54 mm	C7434502	
J11	—	Coaxial jack, ext. clock	—	C20415804	SMA jack; board silk reads "BNC"
JP1	—	Jumper header 2-pin	2.54 mm	C86471	
JP2–JP7	—	Jumper header 3-pin	2.54 mm	C49257	
R1–R5, R9	10 kΩ	Resistor	0805	C17414	
R6, R8, R16, R17, R18	100 Ω	Reset LEDs (R6 / R16 – R18) · CW clock input (R8)	0805	C17408	
R7	0.01 Ω	Current-sense shunt	1206	C105362	
R10–R15	560 Ω	LED resistors	0805	C25319	
R20	1 kΩ	Multi-turn trimmer (Vcore adjust)	—	C57089	
R21	8.2 kΩ	Vcore feedback divider	0805	C17828	
R22	20 Ω	Oscillator series resistor	0805	C17544	
S1	—	DIP switch, 7-position	THT	C331508	
SW1	—	Tactile push-button (B_RST)	5×5 SMD	TBC	part number to be confirmed before ordering
U1	—	DIP-28 IC socket (PROACT)	DIP-28	C72121	
U2	—	TPS74801DRCR 1.5 A adjustable LDO	VSON-10	C105263	
Y1	50 MHz	Clock oscillator, 3225	3225	C2682781	verify before ordering — this position requires an active 3.3 V oscillator

How this document was produced

Every signal name, pin number, and routing option in this document was **cross-checked against the actual board data**, not just transcribed:

- **Netlist** extracted from the final Altium schematic (`Sheet1.SchDoc`) by parsing pins, wires, junctions, net labels and power ports, then rebuilding the connectivity.
- **Chip signal names** from the PROACT pin map (`pins.png`) — the authority when sources disagree.
- **Component positions** in the placement views read directly from the final `PCB1.PcbDoc` (`Components6` stream).
- **BOM** from the final `bom.xlsx` export.
- **CW308 pin functions** from the ChipWhisperer CW308 UFO documentation.

Diagram sources (editable SVG) live in `docs/img/`: `architecture.svg`, `proact_pinout_v2.svg`, `clock_tree.svg`, `power_path.svg`, `board_v2_placement.svg`, `board_v2_silkscreen.svg`, and the per-jumper cards in `docs/img/jumpers/`. The generator scripts are in `docs/tools/`.